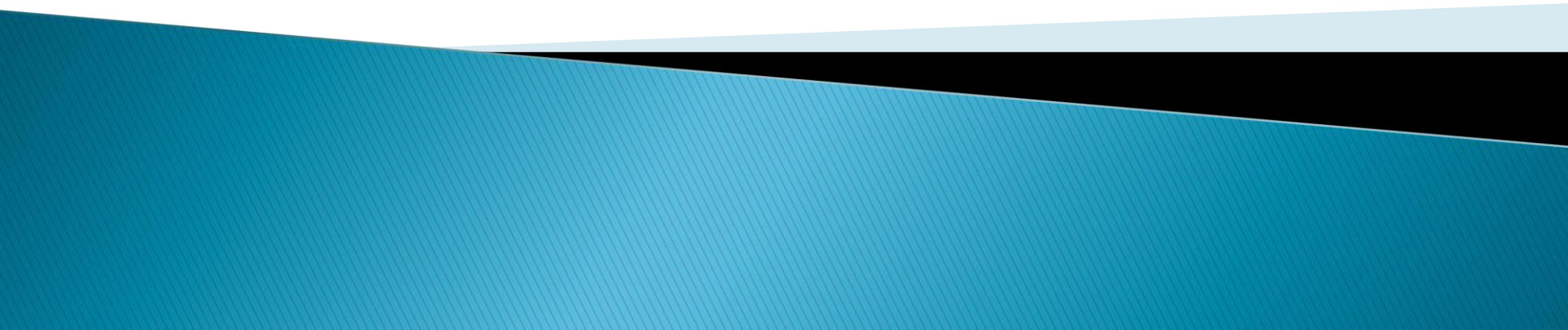
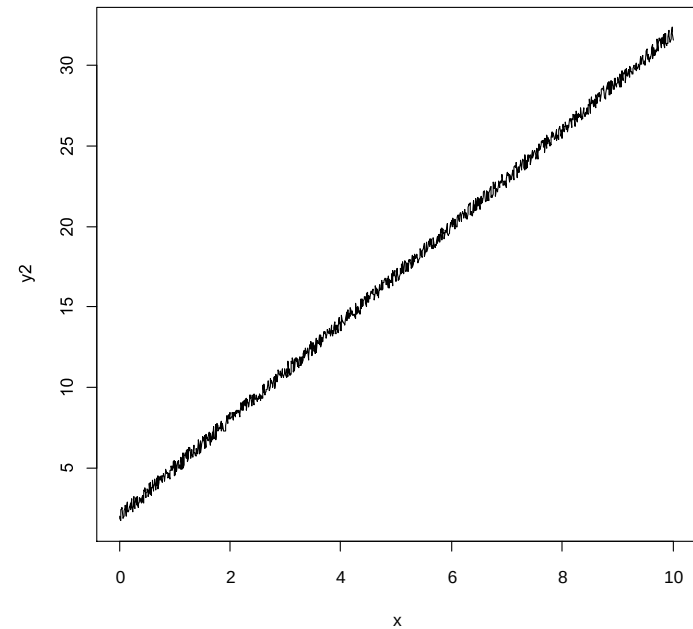
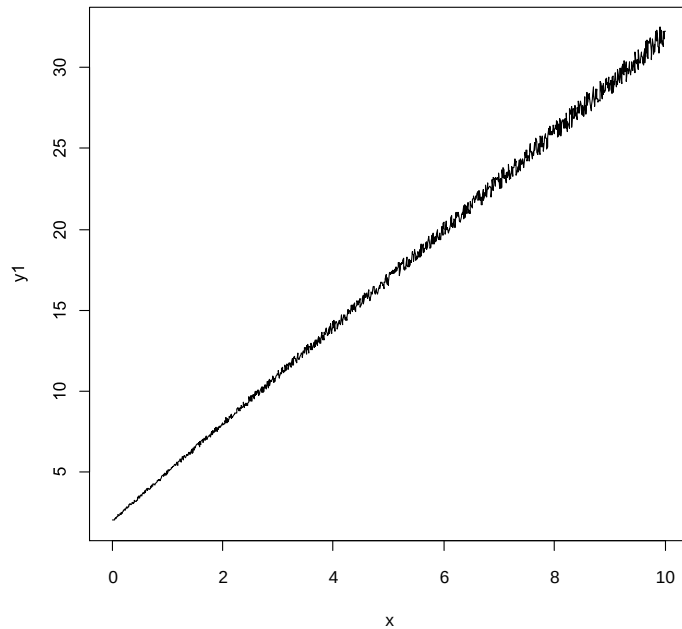


# Bayes Avanzado

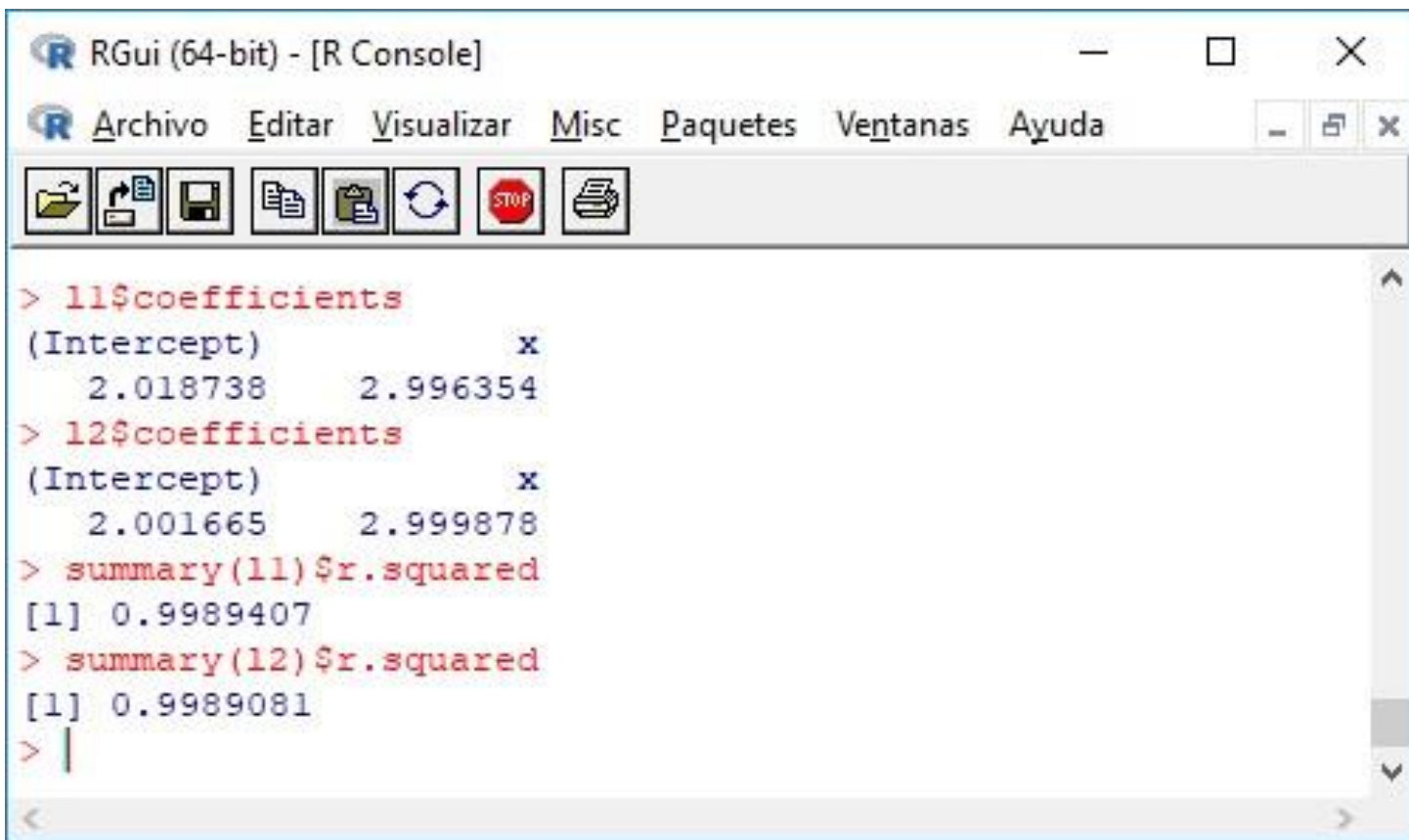
## Agenda

- Regresión lineal bayesiana
  - Regresión logística
- 

- Regresión logística bayesiana
- Redes bayesianas



# Regresión lineal bayesiana



The screenshot shows the RGui (64-bit) - [R Console] window. The menu bar includes Archivo, Editar, Visualizar, Misc, Paquetes, Ventanas, and Ayuda. The toolbar contains icons for file operations and execution. The console displays the following R commands and their output:

```
> l1$coefficients
(Intercept)          x
  2.018738      2.996354
> l2$coefficients
(Intercept)          x
  2.001665      2.999878
> summary(l1)$r.squared
[1] 0.9989407
> summary(l2)$r.squared
[1] 0.9989081
> |
```

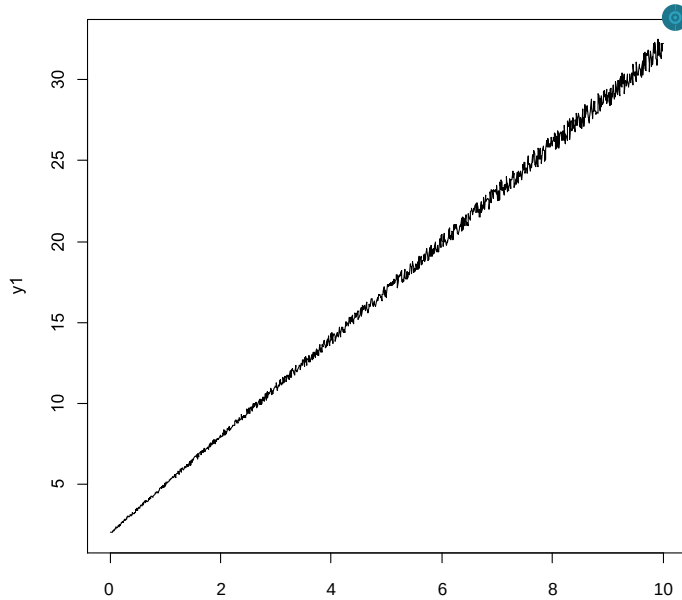
# Regresión lineal bayesiana



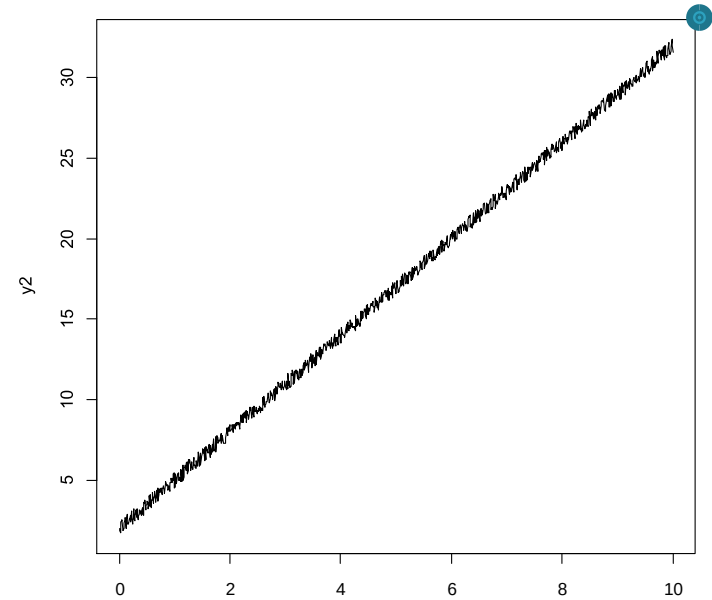
```
> l1$coefficients
(Intercept)          x
    2.018738    2.996354
> l2$coefficients
(Intercept)          x
    2.001780    2.999878
> summary(l1)$r.squared
[1] 0.9989407
> summary(l2)$r.squared
[1] 0.9989081
> |
```

# Regresión lineal bayesiana

Este puede ser normal



Este es outlier



# Regresión lineal bayesiana

x

x

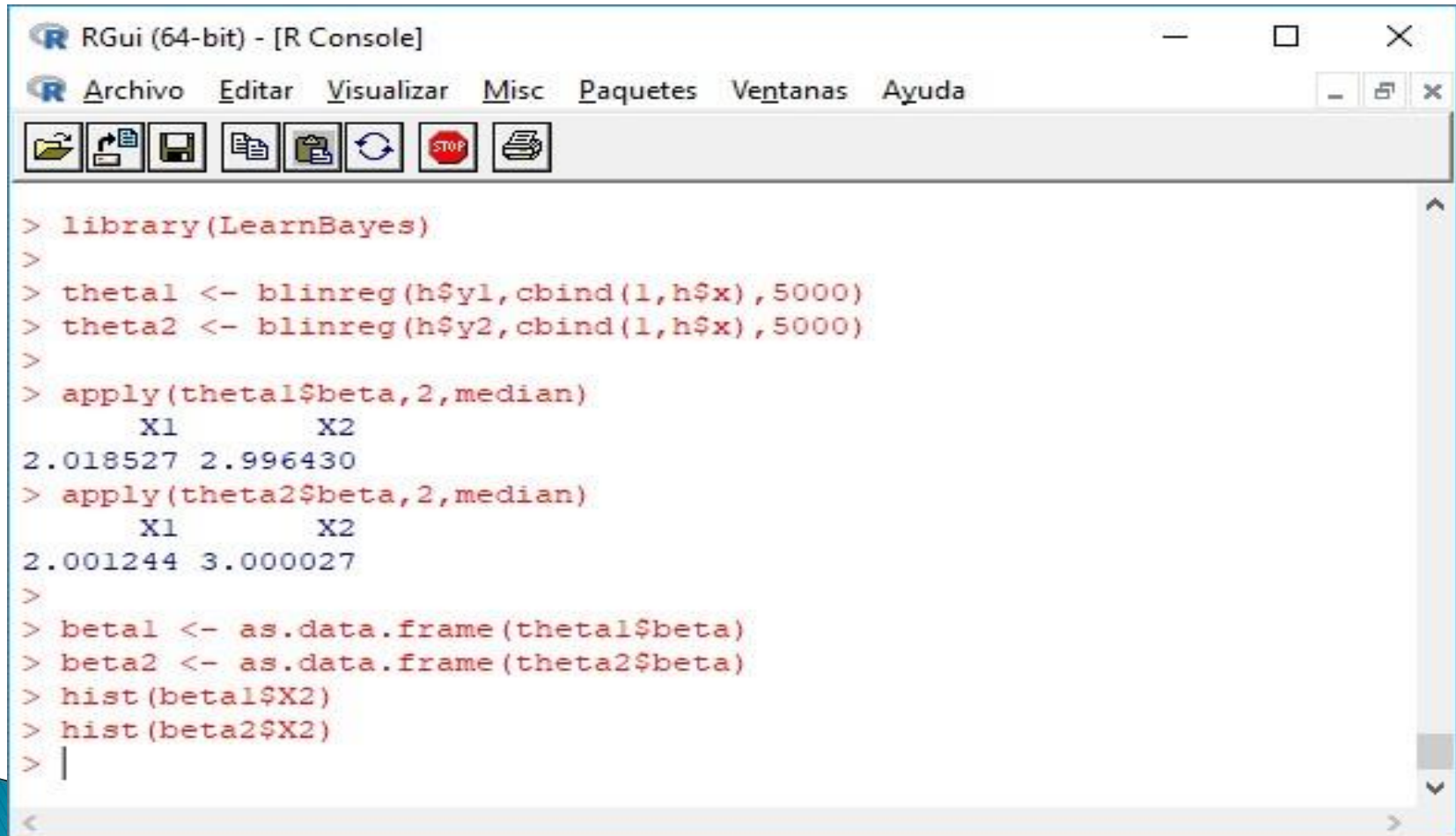
# Regresión lineal bayesiana



# Ahora vamos a LearnBayes



# Ahora vamos a

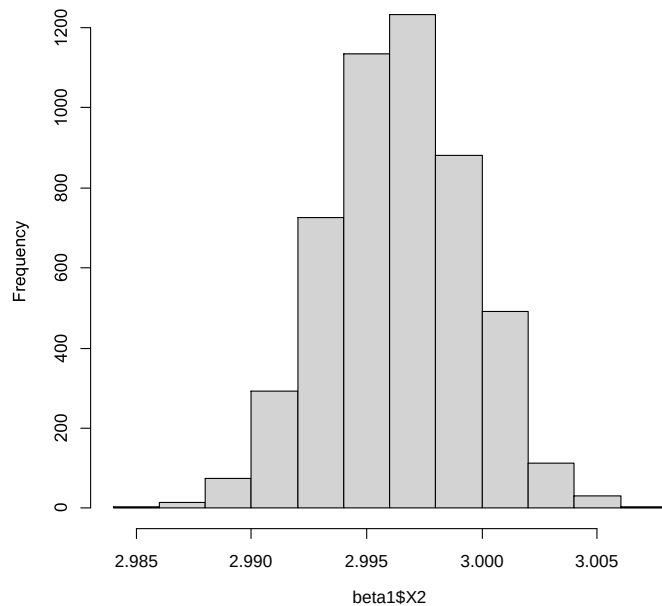


The screenshot shows the RGui (64-bit) - [R Console] window. The menu bar includes Archivo, Editar, Visualizar, Misc, Paquetes, Ventanas, and Ayuda. The toolbar contains icons for file operations and execution. The console displays the following R code and its output:

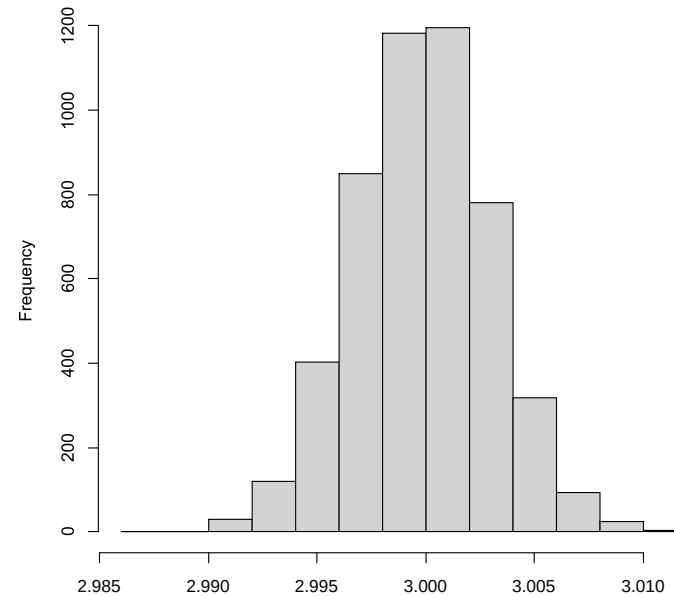
```
> library(LearnBayes)
>
> theta1 <- blinreg(h$y1, cbind(1, h$x), 5000)
> theta2 <- blinreg(h$y2, cbind(1, h$x), 5000)
>
> apply(theta1$beta, 2, median)
      X1      X2
2.018527 2.996430
> apply(theta2$beta, 2, median)
      X1      X2
2.001244 3.000027
>
> betal <- as.data.frame(theta1$beta)
> beta2 <- as.data.frame(theta2$beta)
> hist(betal$X2)
> hist(beta2$X2)
> |
```

# Ahora vamos a LearnBayes

Histogram of beta1\$X2



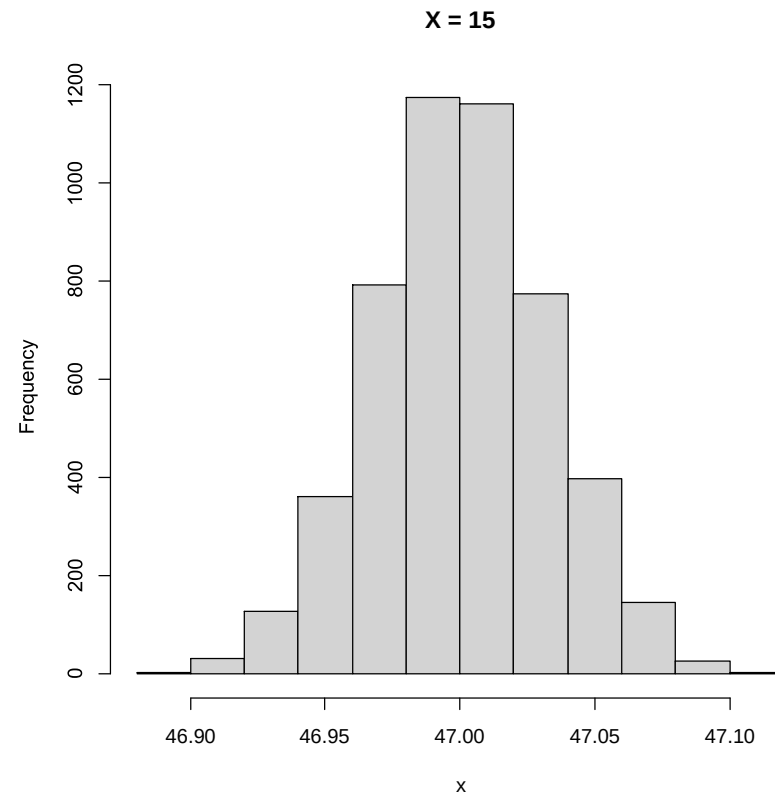
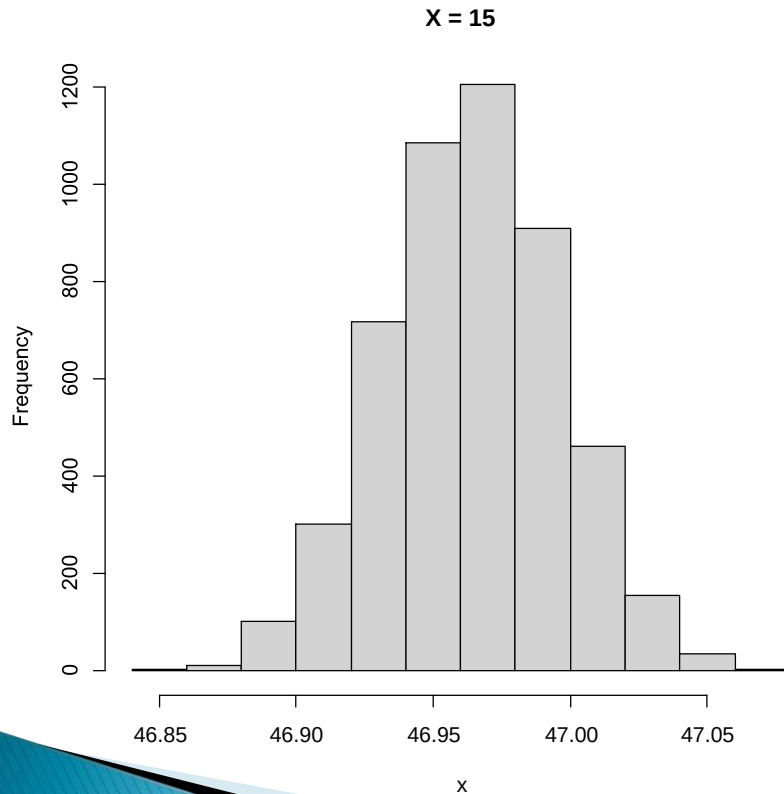
Histogram of beta2\$X2

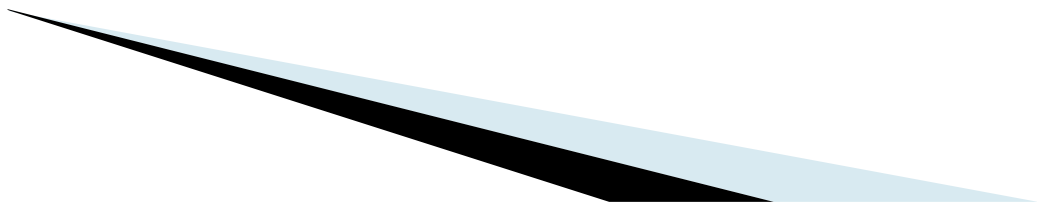


# Ahora vamos a LearnBayes

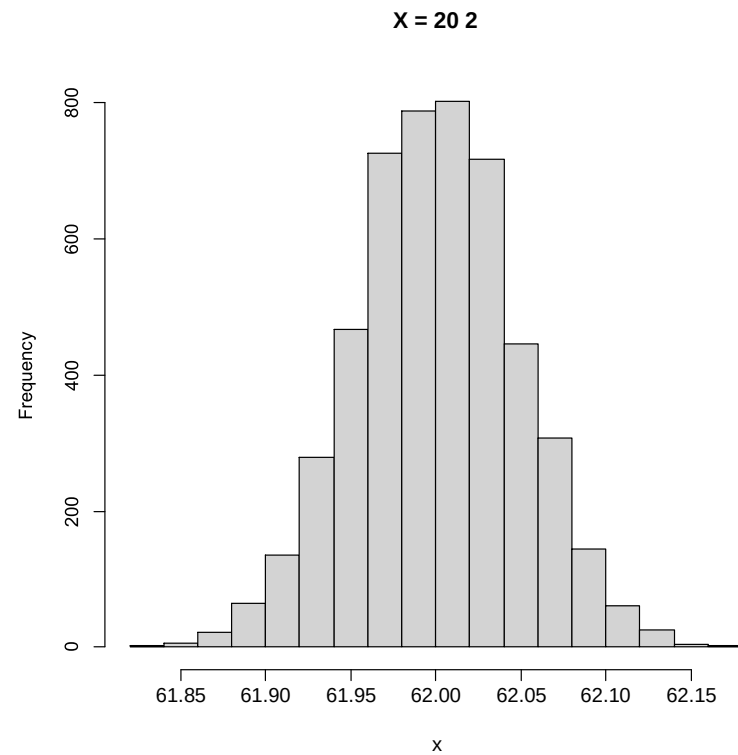
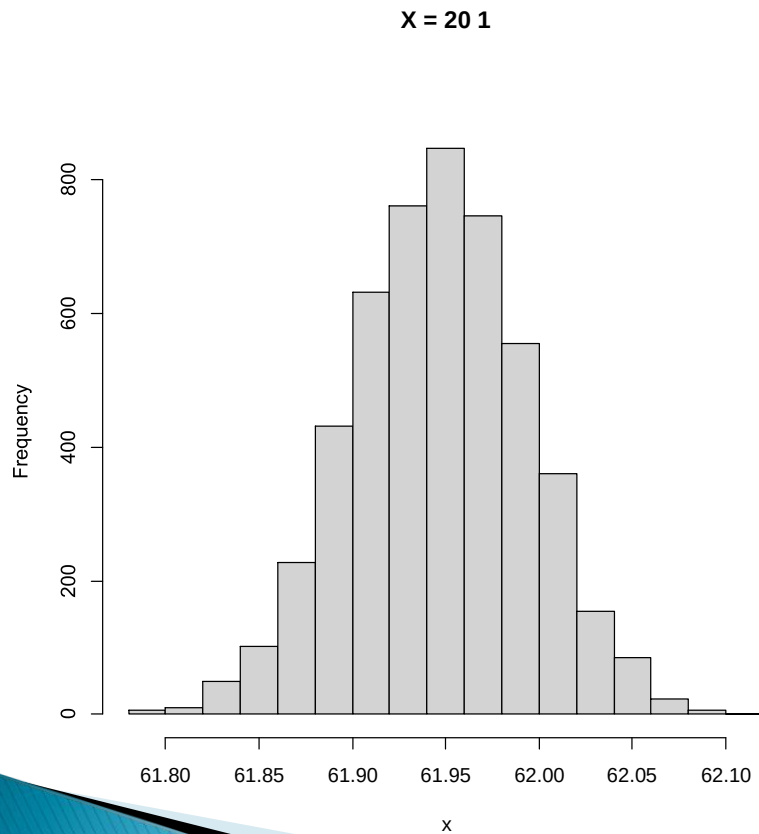
beta2\$X2

# Vamos a extrapolar para $X=15$

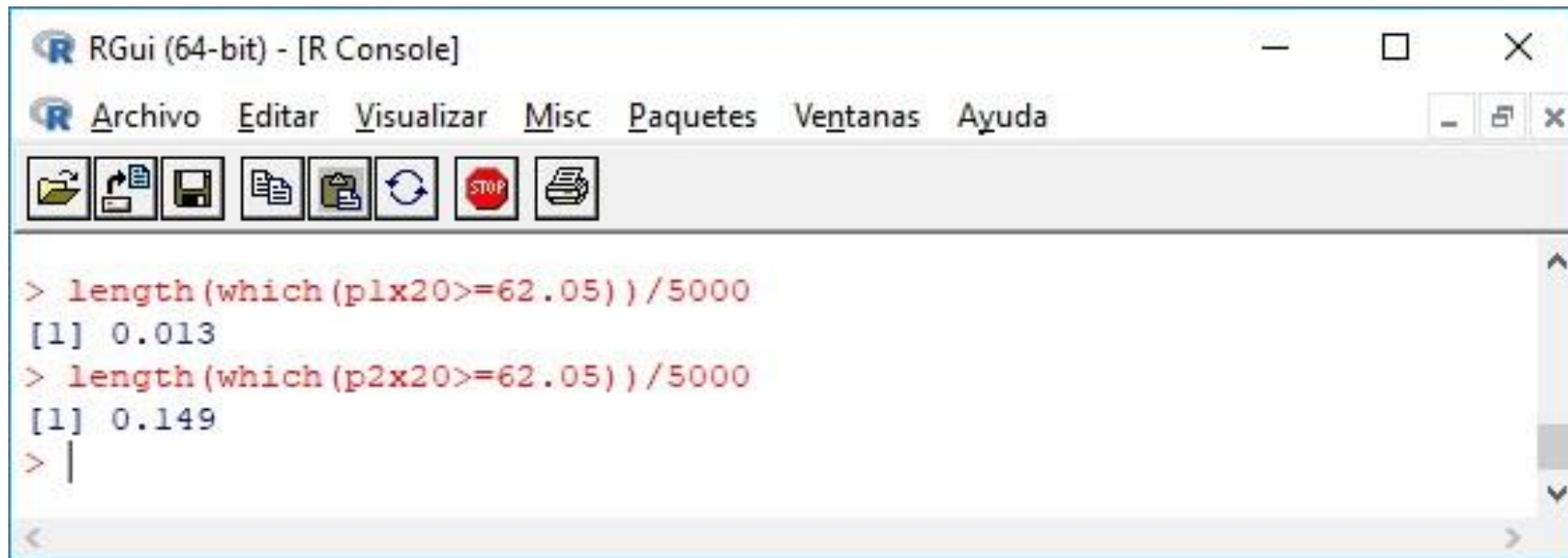




# Vamos a extrapolar para $X=20$

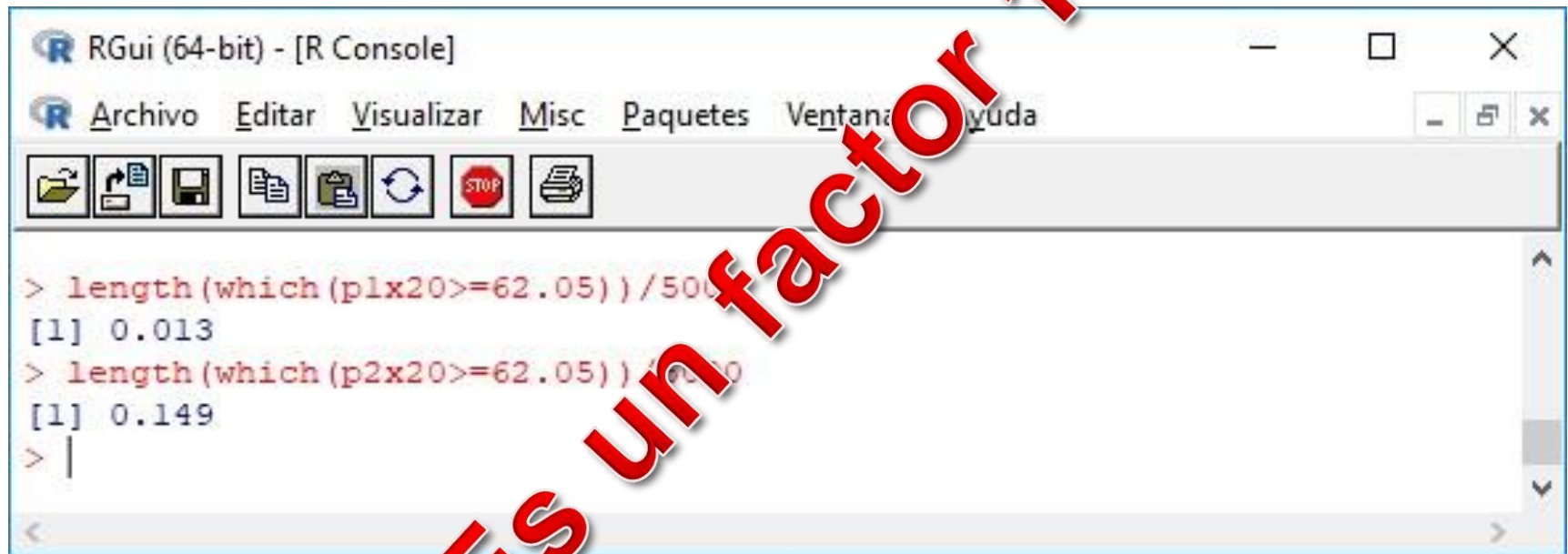


# La diferencia parece pequeña, pero...



```
> length(which(plx20>=62.05))/5000
[1] 0.013
> length(which(p2x20>=62.05))/5000
[1] 0.149
> |
```

# La diferencia parece pequeña, pero...



```
> length(which(plx20 >= 62.05)) / 500
[1] 0.013
> length(which(p2x20 >= 62.05)) / 500
[1] 0.149
> |
```

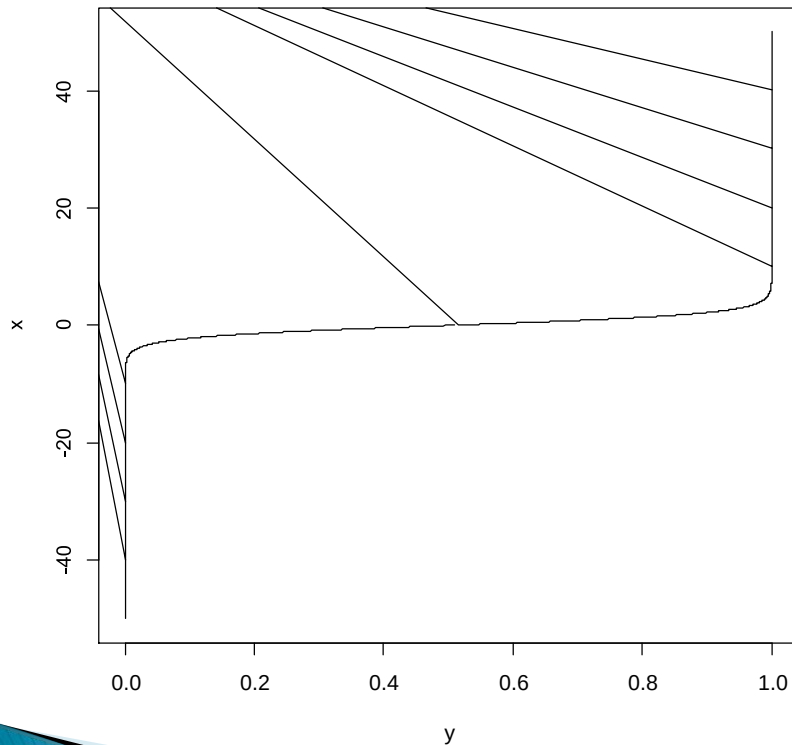


# Regresión logística

- De donde parte:
  - Muestra con atributos conocidos • Categóricos
  - No categóricos • Muestra con resultado conocido
  - Universo con atributos conocidos
- A donde va:
  - Predicción del resultado para el resto del universo
- Cómo trabaja:

# Regresión logística

$$\text{Logit}(P(\text{clase})) = a X + b$$

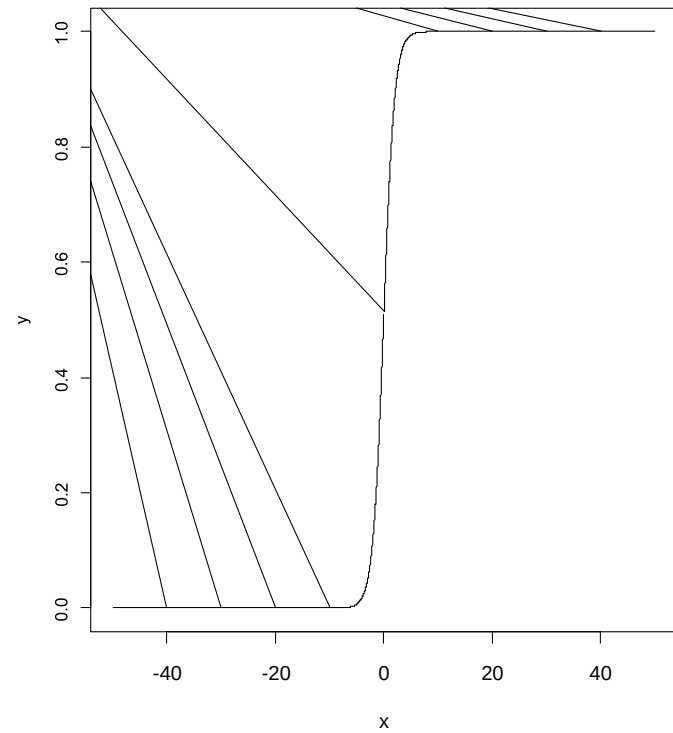


Calcula  $a$  y  $b$  y luego toma la función inversa:

$$P(\text{clase}) = \text{Logist}(a X + b)$$

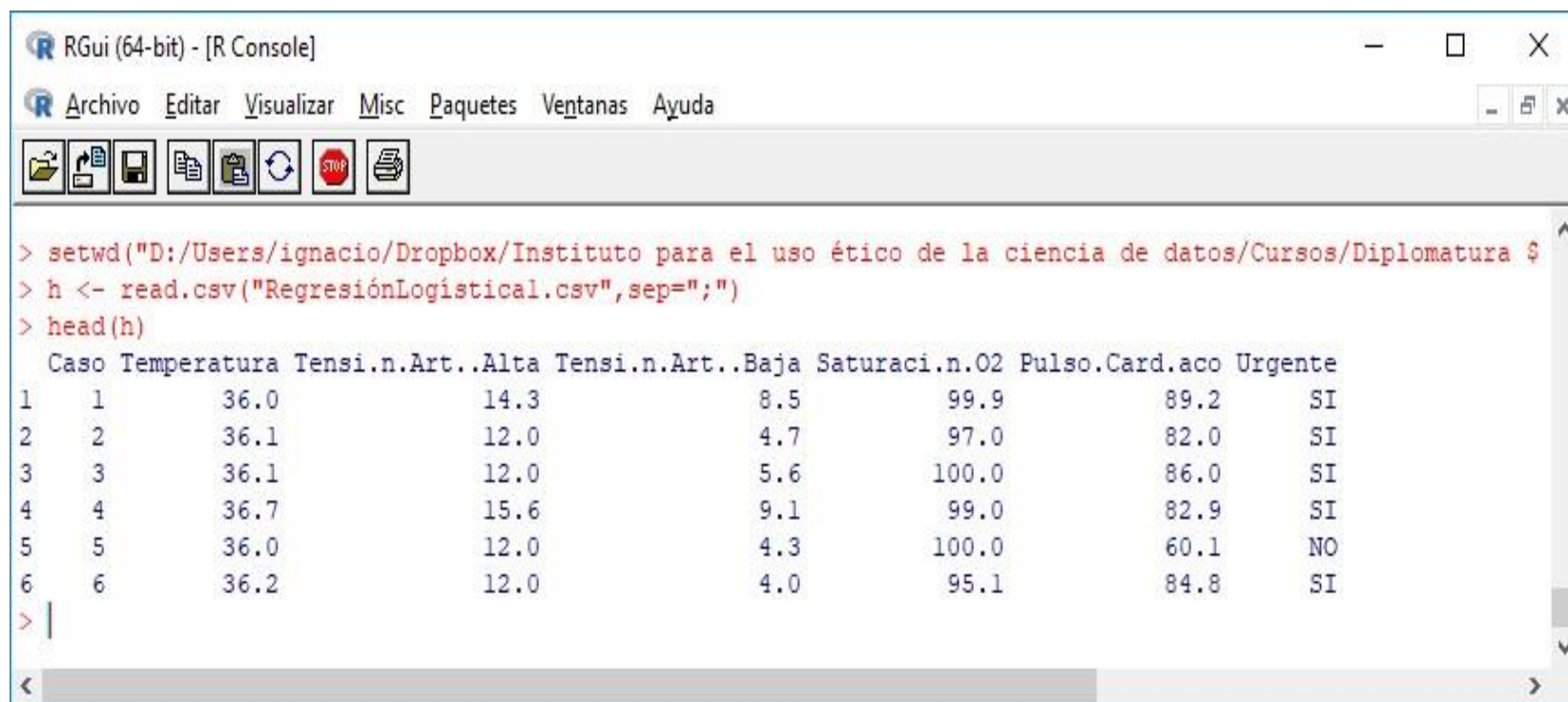
Logit    Logística

# Regresión logística



# Regresión logística

# Regresión logística



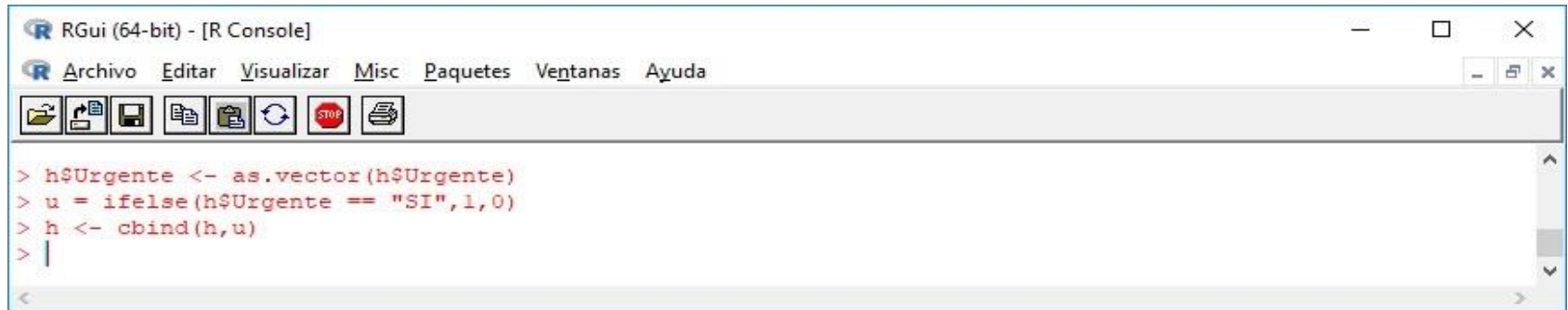
The screenshot shows the RGui (64-bit) - [R Console] window. The menu bar includes Archivo, Editar, Visualizar, Misc, Paquetes, Ventanas, and Ayuda. The toolbar contains icons for file operations and execution. The console displays the following R code and its output:

```
> setwd("D:/Users/ignacio/Dropbox/Instituto para el uso ético de la ciencia de datos/Cursos/Diplomatura $")
> h <- read.csv("RegresiónLogistical.csv", sep=";")
> head(h)
```

|   | Caso | Temperatura | Tensi.n.Art..Alta | Tensi.n.Art..Baja | Saturaci.n.O2 | Pulso.Card.aco | Urgente |
|---|------|-------------|-------------------|-------------------|---------------|----------------|---------|
| 1 | 1    | 36.0        | 14.3              | 8.5               | 99.9          | 89.2           | SI      |
| 2 | 2    | 36.1        | 12.0              | 4.7               | 97.0          | 82.0           | SI      |
| 3 | 3    | 36.1        | 12.0              | 5.6               | 100.0         | 86.0           | SI      |
| 4 | 4    | 36.7        | 15.6              | 9.1               | 99.0          | 82.9           | SI      |
| 5 | 5    | 36.0        | 12.0              | 4.3               | 100.0         | 60.1           | NO      |
| 6 | 6    | 36.2        | 12.0              | 4.0               | 95.1          | 84.8           | SI      |

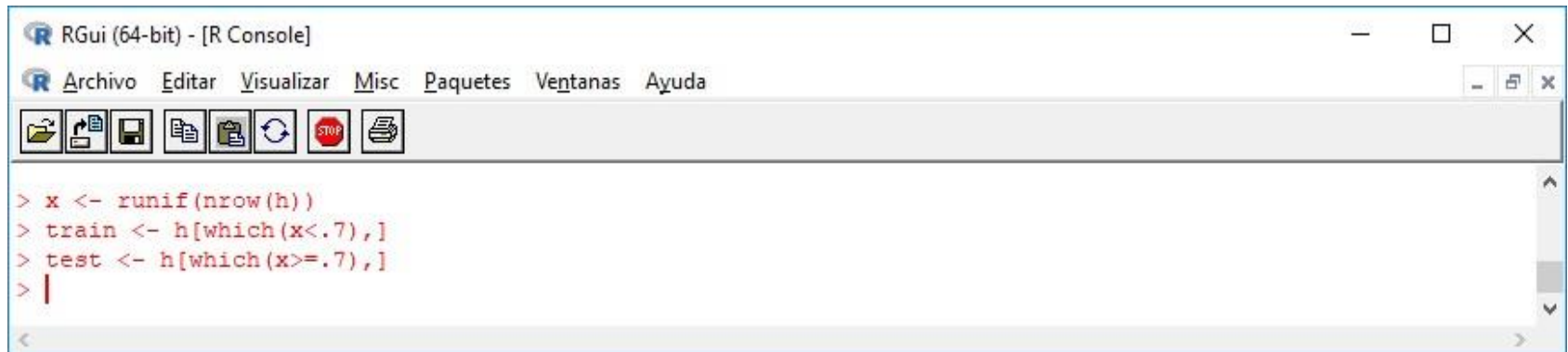
```
> |
```

# Regresión logística



```
RGui (64-bit) - [R Console]
Archivo  Editar  Visualizar  Misc  Paquetes  Ventanas  Ayuda

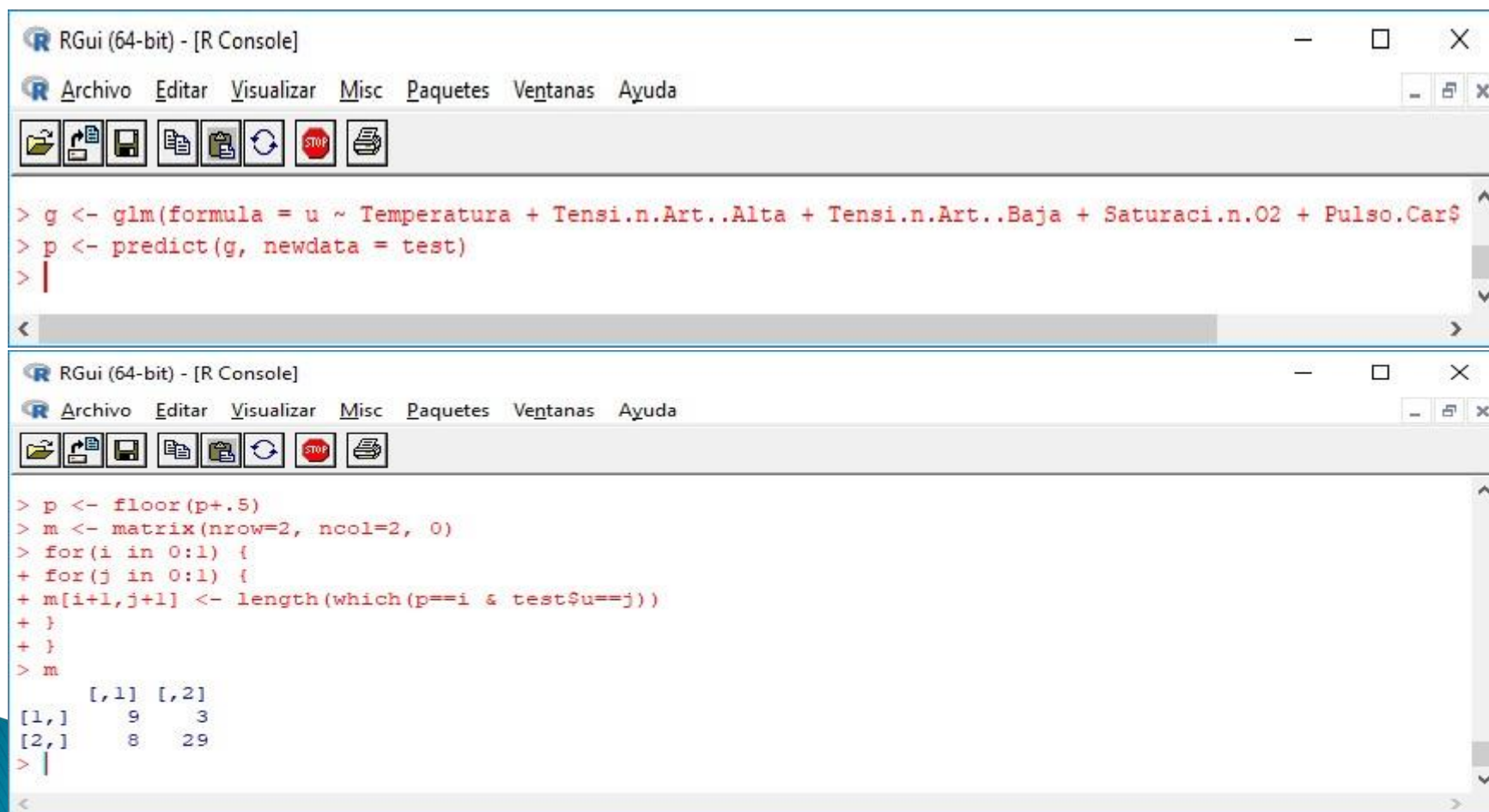
> h$Urgente <- as.vector(h$Urgente)
> u = ifelse(h$Urgente == "SI",1,0)
> h <- cbind(h,u)
> |
```



```
RGui (64-bit) - [R Console]
Archivo  Editar  Visualizar  Misc  Paquetes  Ventanas  Ayuda

> x <- runif(nrow(h))
> train <- h[which(x<.7),]
> test <- h[which(x>=.7),]
> |
```

# Regresión logística



The image shows two screenshots of the RGui (64-bit) - [R Console] window. The first screenshot shows the initial R code for fitting a logistic regression model. The second screenshot shows the code for calculating the predicted probabilities and creating a confusion matrix.

```
> g <- glm(formula = u ~ Temperatura + Tensi.n.Art..Alta + Tensi.n.Art..Baja + Saturaci.n.O2 + Pulso.Car$
> p <- predict(g, newdata = test)
> |
```

```
> p <- floor(p+.5)
> m <- matrix(nrow=2, ncol=2, 0)
> for(i in 0:1) {
+ for(j in 0:1) {
+ m[i+1,j+1] <- length(which(p==i & test$u==j))
+ }
+ }
> m
```

|      | [,1] | [,2] |
|------|------|------|
| [1,] | 9    | 3    |
| [2,] | 8    | 29   |

```
> |
```

# Regresión logística bayesiana

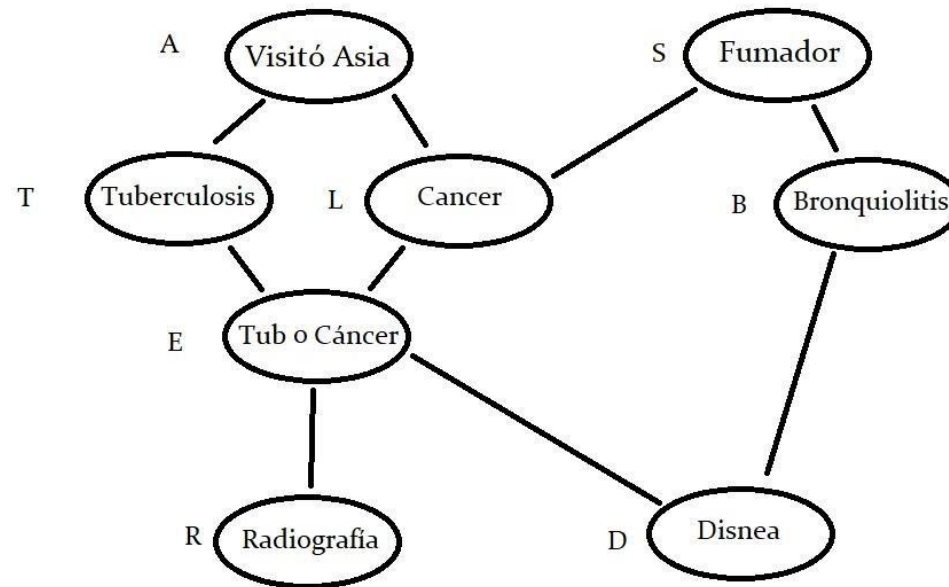
- La utilizamos cuando sospechamos que la distribución de los errores no es aleatoria
- El paquete a utilizar es R2jags.
- Grafo orientado acíclico



# Redes Bayesianas

- Los nodos guardan información probabilística
- Se diseñan en orden causal

# Redes Bayesianas



# Redes Bayesianas

Las probabilidades elementales de cada hecho inicial son:  $P(\text{visitar Asia})$ :  $A = .01$

$P(\text{fumador})$ :  $S = .5$

Pasamos a las probabilidades condicionales:

$P(\text{Tuberculosis/Visitó Asia}) : T = 0.05$

$P(\text{Tuberculosis/No visitó Asia})$ : 0.01

$P(\text{Bronquiolitis/Fumador})$ :  $B = 0.6$

$P(\text{Bronquiolitis/No fumador})$ : 0.3

$P(\text{Cáncer de pulmón/Fumador})$ :  $C = 0.1$   
 $P(\text{Cáncer de pulmón/No Fumador})$ : 0.01

$P(\text{Tub o Cancer/Tub, Cancer}) = 1$

# Redes Bayesianas

$$P(\text{Tub o Cancer} / \text{Tub}, -\text{Cancer}) = 1$$

$$P(\text{Tub o Cancer} / -\text{Tub}, \text{Cancer}) = 1$$

$$P(\text{Tub o Cancer} / -\text{Tub}, -\text{Cancer}) = 0$$

$$P(R/E) = 0.98$$

$$P(R/-E) = 0.05$$

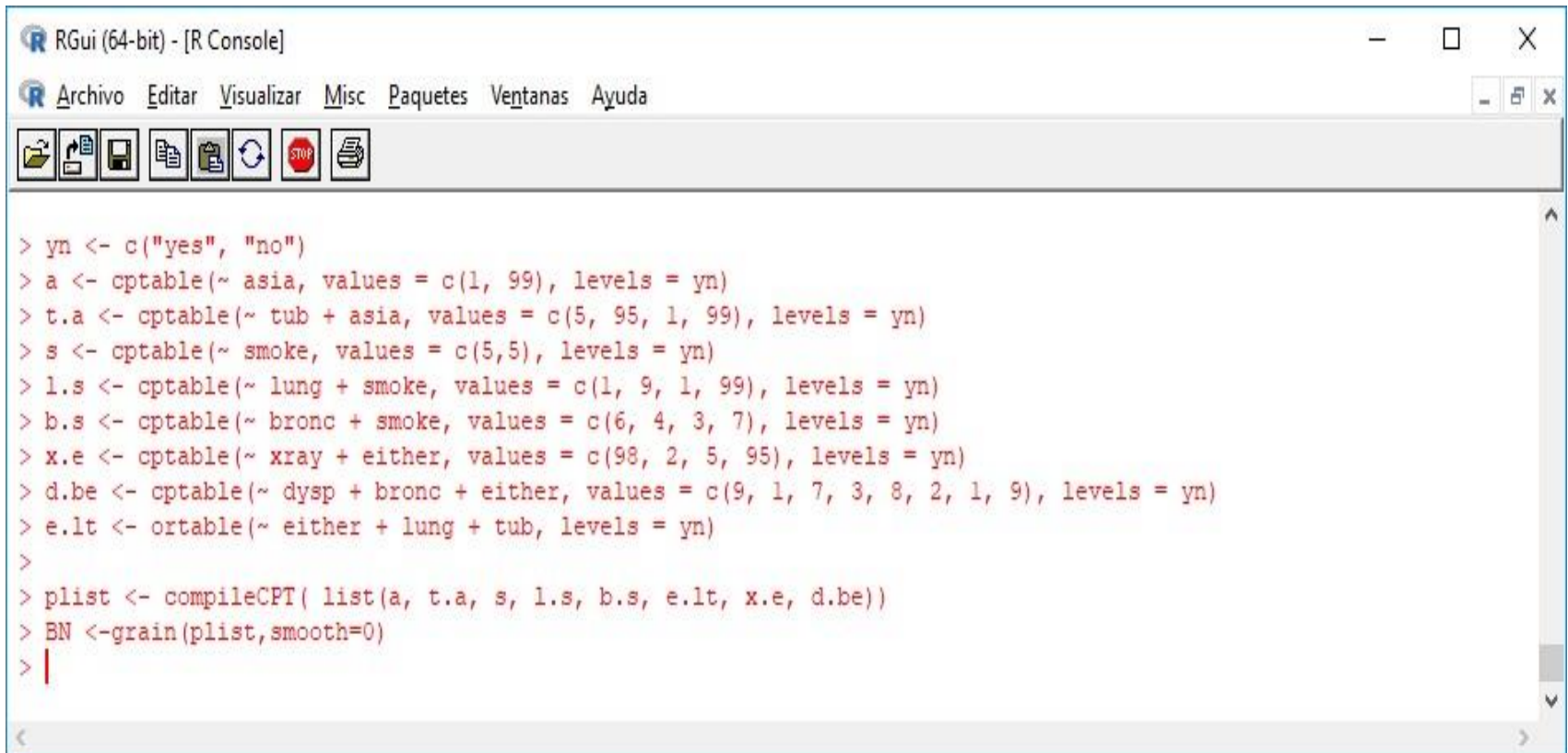
$$P(D/E, B) = 0.90$$

$$P(D/E, -B) = 0.70$$

$$P(D/-E, B) = 0.80$$

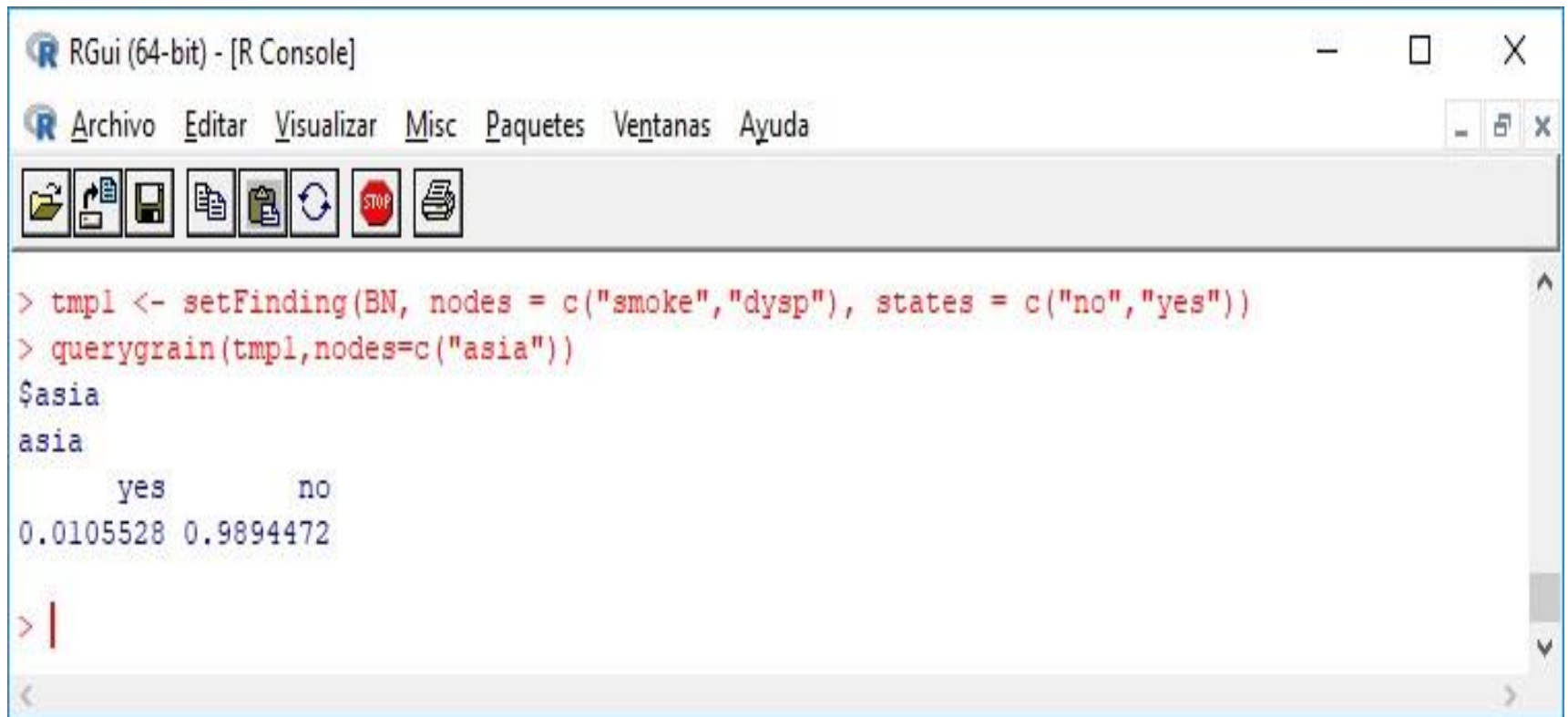
$$P(D/-E, -B) = 0.10$$

# Redes Bayesianas



```
> yn <- c("yes", "no")
> a <- cptable(~ asia, values = c(1, 99), levels = yn)
> t.a <- cptable(~ tub + asia, values = c(5, 95, 1, 99), levels = yn)
> s <- cptable(~ smoke, values = c(5,5), levels = yn)
> l.s <- cptable(~ lung + smoke, values = c(1, 9, 1, 99), levels = yn)
> b.s <- cptable(~ bronc + smoke, values = c(6, 4, 3, 7), levels = yn)
> x.e <- cptable(~ xray + either, values = c(98, 2, 5, 95), levels = yn)
> d.be <- cptable(~ dysp + bronc + either, values = c(9, 1, 7, 3, 8, 2, 1, 9), levels = yn)
> e.lt <- ortable(~ either + lung + tub, levels = yn)
>
> plist <- compileCPT( list(a, t.a, s, l.s, b.s, e.lt, x.e, d.be))
> BN <-grain(plist,smooth=0)
> |
```

# Redes Bayesianas



The screenshot shows the RGui (64-bit) - [R Console] window. The menu bar includes Archivo, Editar, Visualizar, Misc, Paquetes, Ventanas, and Ayuda. The toolbar contains icons for file operations and execution. The console displays the following R code and output:

```
> tmp1 <- setFinding(BN, nodes = c("smoke","dysp"), states = c("no","yes"))
> querygrain(tmp1,nodes=c("asia"))
$asia
asia
      yes      no
0.0105528 0.9894472
> |
```

Muchas Gracias

